THE M_3–$B^{(2)}$ OBSTRUCTION AND THE COSTELLO TCFT IDENTITY

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Abstract. The CY_3 obstruction has three different carriers: the termwise bar-complex operator, the Costello open–closed TCFT operator, and the derived E_1-Hochschild obstruction class. On a strictified cyclic A_∞ model the termwise pair-contraction operator $B^{(2)}_{\mathrm{term}}$ need not commute with m_3. The corrected TCFT operator $B^{(2)}_{\mathrm{TCFT}}$ satisfies

$$\{b, B^{(2)}_{\mathrm{TCFT}}\} = 0, \qquad b = \sum_{k\geq 1} b_k,$$

only after a specific Costello moduli-chain correction datum has been chosen. The derived obstruction vanishes under an explicit spectral sequence hypothesis: the reduced Hochschild cochain filtration is complete, exhaustive, separated, strongly convergent, and has empty total degree -2 line on its first page. Connectivity and unit-connectedness do not replace this degree check. The theorem is an algebraic obstruction statement for the witnessed strictified model; it constructs neither global Φ_3-functoriality, nor $G(X)$, nor compact Hall/CoHA data.

1. Strictified data

Definition 1 (Strictified Hochschild E_1-model)**.** Fix a field of characteristic 0. A strictified Hochschild E_1-model for a smooth proper CY_3 category C is a strictly unital E_1-algebra A in cochain complexes, together with an E_1-quasi-isomorphism from the Hochschild chain object of C to A. It is connective and unit-connected when

$$A^i = 0 \quad (i < 0), \qquad H^0(A) = k,$$

and the unit splitting $A = k \oplus \overline{A}$ is fixed. This strictification is part of the input. It is not a consequence of smoothness and properness alone.

Definition 2 (The framed CY_3 locus)**.** The witnessed framed locus consists of a smooth proper category C equipped with a negative-cyclic CY_3 trace class and a chain-level S^3-framing homotopy on the cyclic bar complex, compatible with the BV operator and the Costello TCFT correction datum below. The specialisation datum (Σ_2, C) used in $\Phi_3^{(\Sigma_2,C)}$ is additional data.

Definition 3 (Termwise and TCFT second operators)**.** Let A be a cyclic A_∞-algebra with Hochschild differential

$$b = \sum_{k\geq 1} b_k.$$

The operator $B^{(2)}_{\mathrm{term}}$ is the bar-desuspended pair contraction built directly from the chosen cyclic pairing on A. It contains no moduli-space correction term. For the strict witness below the normalized terminal-slot convention is used: on a word $[u_1|\cdots|u_r|v]$, $B^{(2)}_{\mathrm{term}}$ contracts the terminal input v with one earlier input u_i, sums over all such i, and carries the Koszul sign obtained after desuspending the bar variables. This fixes the displayed signs; reversing the global orientation of this operator multiplies both sides of the witness by the same sign and does not affect nonvanishing. For the witness terms below the shifted CY_3

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pairing on $s^{-1}A$ has odd degree, and the terminal contraction of $\bar{v} = s^{-1}v$ with $\bar{u}_i = s^{-1}u_i$ is normalized with sign

$$(-1)^{(|\bar{v}|+1)(|\bar{u}_{i+1}|+\cdots+|\bar{u}_r|)}.$$

The operator $B^{(2)}_{\mathrm{TCFT}}$ is the image, under a chosen open–closed TCFT chain map, of the Costello genus-change chain together with its boundary corrections. A TCFT correction datum consists of oriented compactified moduli chains with the Costello signs, a chain-level open–closed TCFT functor to $\mathrm{End}(C_\bullet(A))$, and a representative whose non-principal boundary faces have been absorbed into $B^{(2)}_{\mathrm{TCFT}}$. No equality $B^{(2)}_{\mathrm{term}} = B^{(2)}_{\mathrm{TCFT}}$ is included in this datum. Such an equality requires a separate strictifying homotopy.

2. Termwise failure

Proposition 4 (A strict nonzero witness)**.** *There is a cyclic CY_3 A_∞-model with nonzero m_3 for which the termwise graded commutator*

$$[m_3, B^{(2)}_{\mathrm{term}}]_{\mathrm{gr}} = m_3 B^{(2)}_{\mathrm{term}} - B^{(2)}_{\mathrm{term}} m_3$$

is nonzero. Take the graded vector space

$$A = \langle e, a, b, w\rangle, \qquad |e| = 0,\ |a| = 1,\ |b| = 2,\ |w| = 3,$$

with cyclic CY_3 pairing

$$\langle e, w\rangle = 1, \qquad \langle a, b\rangle = 1$$

up to the usual Koszul symmetry, and minimal strictly unital A_∞-structure whose only nonzero higher product on the augmentation ideal is

$$m_3(a, a, a) = \alpha b, \qquad \alpha \neq 0.$$

For the bar word $x = [a|a|a|a|b]$, the bar-desuspended convention gives

$$B^{(2)}_{\mathrm{term}} x = 4[a|a|a], \qquad m_3 B^{(2)}_{\mathrm{term}} x = 4\alpha[b],$$

and

$$m_3 x = \alpha([b|a|b] + [a|b|b]), \qquad B^{(2)}_{\mathrm{term}} m_3 x = 2\alpha[b].$$

Hence

$$[m_3, B^{(2)}_{\mathrm{term}}]_{\mathrm{gr}} x = 2\alpha[b] \neq 0.$$

Proof. The only Stasheff identities to check are those containing nested m_3-terms, since $m_1 = 0$ and m_2 vanishes on the augmentation ideal. Every nested m_3-term has a b-input after the first application, and m_3 is zero on such inputs. Strict unitality kills all terms containing e. Thus the displayed operations define a minimal cyclic A_∞-model.

The trace displayed in the statement is the direct pair-contraction calculation. Write $\bar{a} = s^{-1}a$, $\bar{b} = s^{-1}b$. Then $|\bar{a}| = 0$ and $|\bar{b}| = 1$, so moving the terminal $\bar{b}$ past any string of $\bar{a}$'s contributes no Koszul sign. The four terminal-slot pairings of b with one of the four a's therefore give $4[a|a|a]$. The two m_3-insertions not containing the terminal b give $[b|a|b]$ and $[a|b|b]$. In each of these two words the terminal b contracts with the unique earlier a. The first has no intervening factor between a and the terminal b. In the second, the sign exponent is

$$(|\bar{b}| + 1)|\bar{b}| = (1 + 1) \cdot 1,$$

hence it is even. Therefore both contractions are positive, and $B^{(2)}_{\mathrm{term}} m_3 x = 2\alpha[b]$. Since the field has characteristic 0, $2\alpha[b]$ is nonzero. $\square$

Proposition 5 (Local $\mathbb{P}^2$ is only a noncompact guide). *The calculation above is an algebraic strict-model witness for a non-formal m_3–$B^{(2)}_{\mathrm{term}}$ failure. The toric threefold* $\mathrm{Tot}(\mathcal{O}_{\mathbb{P}^2}(-3))$ *exhibits the same non-formal mechanism in a noncompact CY_3 geometry, but the displayed four-generator witness is not by itself a compact CY_3 construction.*

Proof. Local $\mathbb{P}^2$ is noncompact and toric. Its role in this calculation is diagnostic: a nonzero m_3 can make the termwise $B^{(2)}_{\mathrm{term}}$ fail on a strict bar model. The compact smooth-proper hypotheses, the specialisation datum (Σ_2, C), a global Φ_3 construction, and compact Hall correspondences are not present in that diagnostic. □

Proposition 6 (Toric positive-half comparison is orthogonal). *On $\mathbb{C}^3$, and on finite toric CY_3 atlases equipped with simplex-compatible DWR/Ran Rees hCS–Hall data, the Hall comparison exists at finite Rees positive-half level:*

$$\mathrm{CoHA}^{\mathrm{sph}}_{\mathrm{crit},T}(\mathbb{C}^3) \cong Y^+(\widehat{\mathfrak{gl}}_1).$$

This positive half is not $\mathcal{W}_{1+\infty}$, and it does not remove the m_3–$B^{(2)}_{\mathrm{term}}$ failure above. Compact critical CoHA, quasi-NCCR, Hall–Drinfeld double, and global Φ_3-functoriality statements remain conditional on their own orientation, compact-support, pairing, completion, and PBW data.

Proof. The finite toric comparison is a Hall statement about critical correspondences and finite Rees hCS observables. The obstruction above is a bar-complex statement about a chosen strictified Hochschild E_1-model. The two live in different complexes. The Schiffmann–Vasserot identification supplies the $\mathbb{C}^3$ positive half, and finite toric gluing assembles such positive halves when the cyclic atlas and face maps are supplied. None of those data changes the explicit bar word $[a|a|a|a|b]$ or the nonzero commutator $[m_3, B^{(2)}_{\mathrm{term}}]_{\mathrm{gr}} x = 2\alpha[b]$. □

3. TCFT cancellation

Theorem 7 (Costello TCFT identity). *Let A be a cyclic A_∞-algebra equipped with a Costello open–closed TCFT correction datum as above. For the corrected operator $B^{(2)}_{\mathrm{TCFT}}$,*

$$\{b, B^{(2)}_{\mathrm{TCFT}}\} = b \circ B^{(2)}_{\mathrm{TCFT}} + B^{(2)}_{\mathrm{TCFT}} \circ b = 0.$$

Proof. Costello's TCFT theorem identifies cyclic Calabi–Yau A_∞-data with open TCFT data and supplies the universal open–closed extension whose closed sector is Hochschild homology. Under the chosen correction datum, the oriented boundary relation in the relevant compactified moduli-chain complex has two principal faces, mapping to

$$b \circ B^{(2)}_{\mathrm{TCFT}} \quad \text{and} \quad B^{(2)}_{\mathrm{TCFT}} \circ b.$$

The non-principal faces are exactly the terms absorbed into the corrected representative $B^{(2)}_{\mathrm{TCFT}}$. Applying the TCFT chain map to that boundary relation gives the displayed total anticommutator. The Costello orientation convention is part of the datum: the two principal boundary orientations are the Koszul-signed summands of the graded anticommutator, which in the present cohomological convention is $bB^{(2)}_{\mathrm{TCFT}} + B^{(2)}_{\mathrm{TCFT}} b$. The proof uses the total Hochschild differential $b = \sum_k b_k$; it does not assert $\{b_k, B^{(2)}_{\mathrm{TCFT}}\} = 0$ for each k. □

4. Filtered obstruction

Theorem 8 (Derived E_1-Hochschild vanishing). *Let C lie in the witnessed framed locus and let A be a connective unit-connected strictified Hochschild E_1-model for C. Assume the reduced Hochschild cochain complex*

$$C^{\bullet}_{E_1}(A, A)$$

with its bar-length filtration is complete, exhaustive, separated, and strongly convergent to $\mathrm{HH}^{\bullet}_{E_1}(A, A)$. *Assume its first page is*

$$E_1^{p,q} = \mathrm{Hom}\big((s\overline{A})^{\otimes p}, A\big)^q$$

and satisfies

$$E_1^{p,q} = 0 \qquad \text{whenever } p + q = -2.$$

Fix a chain-level comparison map from the S^3-framing obstruction complex to $C^{\bullet}_{E_1}(A, A)[2]$, and assume the resulting obstruction cocycle has target total degree -2. Then

$$\mathrm{HH}^{-2}_{E_1}(A, A) = 0.$$

Consequently that derived m_3–$B^{(2)}$ framing obstruction vanishes as a class in $\mathrm{HH}^{-2}_{E_1}(A, A)$.

Proof. The total degree -2 line on the first page is empty by hypothesis. Every later page is a subquotient of the previous page on the same total line, so

$$E_\infty^{p,q} = 0 \qquad (p + q = -2).$$

Strong convergence identifies $E_\infty^{p,-2-p}$ with the associated graded pieces of the induced filtration on $\mathrm{HH}^{-2}_{E_1}(A, A)$. Since all associated graded pieces are zero and the filtration is separated and complete, the filtered group itself is zero. Thus any obstruction class whose comparison target is $\mathrm{HH}^{-2}_{E_1}(A, A)$ vanishes. □

Remark 9. The hypotheses above are the theorem. Connectivity and $H^0(A) = k$ are not by themselves a proof of the empty total degree -2 line. A compact non-formal model outside this filtration locus requires an independent strictification theorem, a formality theorem transporting it to such a model, or a Goodwillie convergence theorem with the relevant derived limit terms killed.

5. No global construction

Definition 10. For a fixed strictified model A, the TCFT obstruction datum is

$$\mathfrak{o}_{\mathrm{TCFT}}(A) = \Big(\{b, B^{(2)}_{\mathrm{TCFT}}\}, [\mathrm{Obs}_{S^3}(A)]\Big) \in \mathrm{End}(C_\bullet(A)) \oplus \mathrm{HH}^{-2}_{E_1}(A, A).$$

It records the total TCFT anticommutator and the derived S^3-framing class on the chosen strictified Hochschild model.

Proposition 11. *The vanishing $\mathfrak{o}_{\mathrm{TCFT}}(A) = 0$ gives only the algebraic framing-obstruction statement for A. It does not construct:*

a functorial global Φ_3,
a global object $G(X)$,
a compact Hall algebra or compact CoHA,
an orientation/Pfaffian line,
a PBW basis or a no-extra-relations theorem.

Proof. The first component of $\mathfrak{o}_{\mathrm{TCFT}}(A)$ is an endomorphism of the Hochschild chain complex. The second component is an E_1-Hochschild cohomology class. Neither component contains a moduli stack of compact objects, vanishing-cycle sheaves, compact extension correspondences, a coproduct, a non-degenerate Hall pairing, a radical quotient, or a filtration whose associated graded algebra is identified with a symmetric algebra. These are source-side structures. The obstruction calculation supplies no map from $\mathfrak{o}_{\mathrm{TCFT}}(A)$ to those data. $\square$

Remark 12. The false strengthening is the termwise identity. The statements

$$[b_k, B^{(2)}_{\mathrm{term}}]_{\mathrm{gr}} = 0, \qquad \{b_k, B^{(2)}_{\mathrm{TCFT}}\} = 0 \quad (k \geq 1)$$

are stronger than the TCFT theorem and are not used. The proved identity is the total corrected one:

$$\{\sum_k b_k,\ B^{(2)}_{\mathrm{TCFT}}\} = 0.$$

References

[1] K. Costello, *Topological conformal field theories and Calabi–Yau categories*, Adv. Math. **210** (2007), 165–214; arXiv:math/0412149.